Experiment No-5

Aim

To write a Python program to solve the Missionaries and Cannibals problem using Breadth-First Search (BFS) in Artificial Intelligence.

Programming Code:

```
from collections import deque

def solve():
    start = (3, 3, 1)
    goal = (0, 0, 0)
    moves = [(1, 0), (2, 0), (0, 1),
             (0, 2), (1, 1)]

    queue = deque([(start, [start])])
    visited = {start}

    while queue:
        (m, c, b), path = queue.popleft()

        if (m, c, b) == goal:
            print("Solution:")
            for state in path:
                print(state)
            return

        for dm, dc in moves:
            if b == 1:
```

```
            nm, nc, nb = m-dm, c-dc, 0
        else:
            nm, nc, nb = m+dm, c+dc, 1

        if not (0 <= nm <= 3 and 0 <= nc <= 3):
            continue

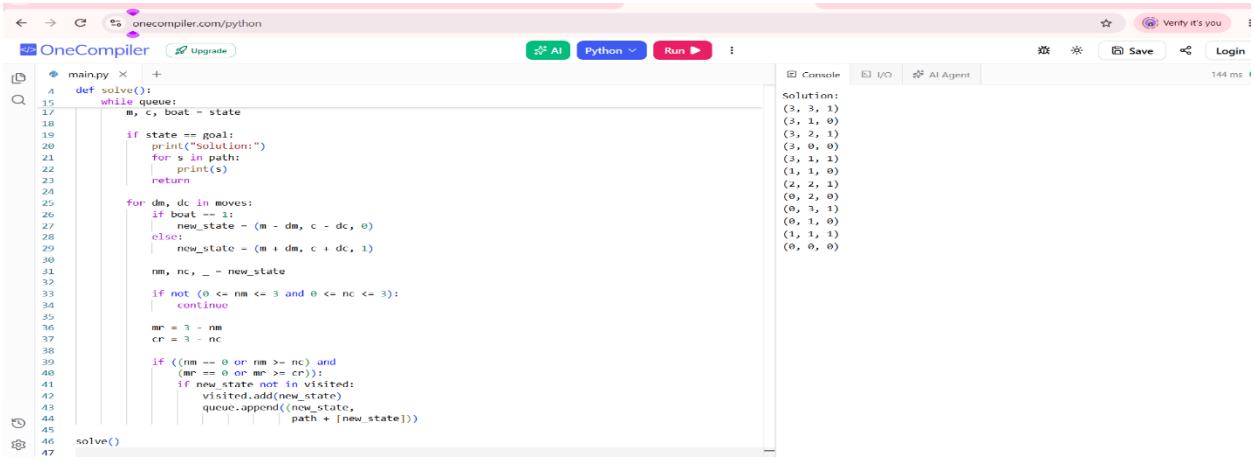
        mr, cr = 3-nm, 3-nc

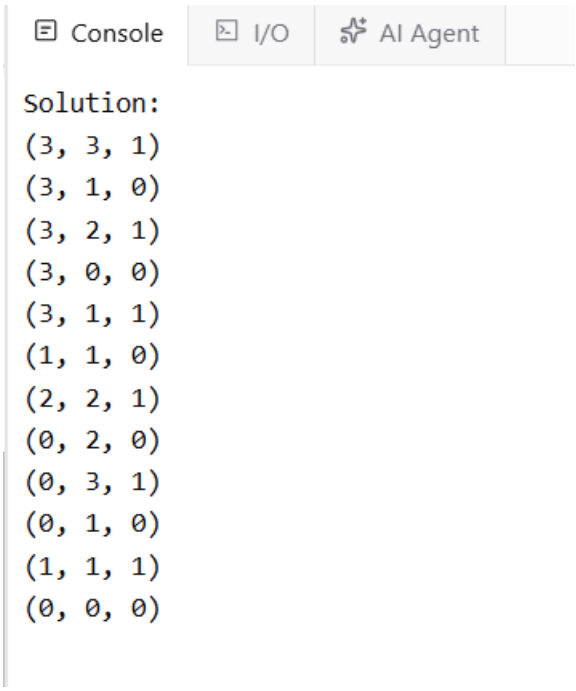
        if ((nm == 0 or nm >= nc) and
            (mr == 0 or mr >= cr)):
            new_state = (nm, nc, nb)

            if new_state not in visited:
                visited.add(new_state)
                queue.append(
                    (new_state, path + [new_state]))

solve()
```

output:





Result

The Missionaries and Cannibals problem was solved successfully using the Breadth-First Search (BFS) algorithm, moving all three missionaries and three cannibals safely from the left bank to the right bank without violating the problem constraints.